

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

June 2025 Reconstructed Past Paper

DETAILED SOLUTIONS

Original question followed by a full worked solution

28 working units 94 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Every working unit appears in exactly the same order as the Student book. The complete source question appears first; the next page gives a detailed worked solution and the final answer.

Unit ownership rule: a question is assigned to Unit 2 when its assessed or terminal statement belongs to Unit 2, even if it uses Unit 1 prerequisites. This book contains only the demands owned by Unit 1. The 200 marks of the two Linear Higher papers were partitioned exactly; this Unit 1 selection contains 94 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

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Working Unit 01

4MA1-1H Q2

ORIGINAL QUESTION**2 marks**

The height of each cup is 9 cm, correct to the nearest cm

(b) Write down the lower bound of the height.

..... cm
(1)

The weight of each cup is 120 g, correct to the nearest 10 g

(c) Write down the upper bound of the weight.

..... g
(1)

Worked Solution 01

Bounds from stated accuracy

UNIT 1
2 marks**Height**

$$9 \text{ cm to the nearest cm} \Rightarrow 8.5 \leq h < 9.5.$$

Weight

$$120 \text{ g to the nearest 10 g} \Rightarrow 115 \leq w < 125.$$

Final answer

$$(b) \ 8.5 \text{ cm} \quad (c) \ 125 \text{ g}$$

Dependency check Independent of the excluded Unit 2 profit part Q2(a).

Working Unit 02

4MA1-1H Q4

ORIGINAL QUESTION

2 marks

4 $x^5 \times x^7 = x^m$

(a) Find the value of m

$$m = \dots\dots\dots$$

(1)

$y^8 \div y^3 = y^n$

(b) Find the value of n

$$n = \dots\dots\dots$$

(1)

Worked Solution 02

Index laws: multiply and divide

UNIT 1
2 marks**Multiply equal bases**

$$x^5 x^7 = x^{5+7} = x^{12} \Rightarrow m = 12.$$

Divide equal bases

$$y^8 \div y^3 = y^{8-3} = y^5 \Rightarrow n = 5.$$

Final answer

$$m = 12, \quad n = 5$$

Working Unit 03

4MA1-1H Q4

ORIGINAL QUESTION**2 marks**

(c) Simplify fully $(5a^4r^2)^3$

(2)

Worked Solution 03

Power of a product

UNIT 1
2 marks**Apply the outer power**

$$(5a^4r^2)^3 = 5^3(a^4)^3(r^2)^3 = 125a^{12}r^6.$$

Final answer

$$125a^{12}r^6$$

Working Unit 04

4MA1-1H Q6

ORIGINAL QUESTION

3 marks

6 (a) Solve $x - 4 = \frac{3 + 2x}{6}$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

Worked Solution 04

Solve a fractional linear equation

UNIT 1
3 marks**Clear the denominator**

$$x - 4 = \frac{3 + 2x}{6} \Rightarrow 6x - 24 = 3 + 2x.$$

Collect and solve

$$4x = 27 \Rightarrow x = \frac{27}{4} = 6.75.$$

Final answer

$$x = \frac{27}{4} = 6.75$$

Working Unit 05

4MA1-1H Q6

ORIGINAL QUESTION**3 marks**

(b) (i) Factorise $y^2 - 11y + 30$

(2)

(ii) Hence solve $y^2 - 11y + 30 = 0$

(1)

Worked Solution 05

Factorise, then solve

UNIT 1
3 marks**Factorise**

$$y^2 - 11y + 30 = (y - 5)(y - 6).$$

Use the zero-product rule

$$(y - 5)(y - 6) = 0 \Rightarrow y = 5 \text{ or } y = 6.$$

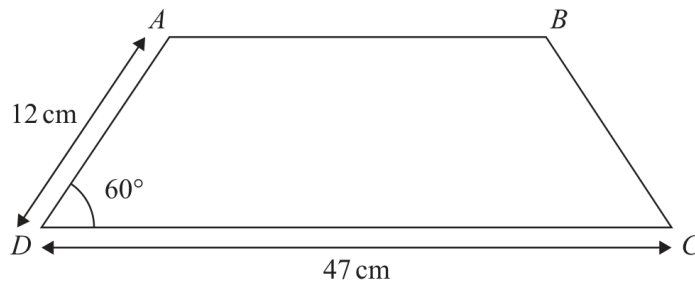
Final answer

$$(y - 5)(y - 6), \quad y = 5 \text{ or } 6$$

Dependency check The 'Hence' solution depends on the factorisation and therefore remains in the same working unit.

Working Unit 06

4MA1-1H Q9

ORIGINAL QUESTION**5 marks****9**Diagram **NOT**
accurately drawn $ABCD$ is a trapezium with one line of symmetry.

$$\text{angle } ADC = 60^\circ \quad AD = 12 \text{ cm} \quad DC = 47 \text{ cm}$$

Work out the area of the trapezium.

Give your answer correct to 3 significant figures.

Show your working clearly.

..... cm^2 **(Total for Question 9 is 5 marks)**

Worked Solution 06

Area of a symmetric trapezium

UNIT 1
5 marks**Height and horizontal projection**

$$h = 12 \sin 60^\circ = 6\sqrt{3}, \quad 12 \cos 60^\circ = 6.$$

Upper base

$$AB = 47 - 6 - 6 = 35 \text{ cm.}$$

Area

$$A = \frac{1}{2}(47 + 35)(6\sqrt{3}) = 246\sqrt{3} = 426.084 \dots$$

Final answer

$$426 \text{ cm}^2 \text{ (3 s.f.)}$$

Dependency check The line of symmetry makes the two horizontal projections equal.

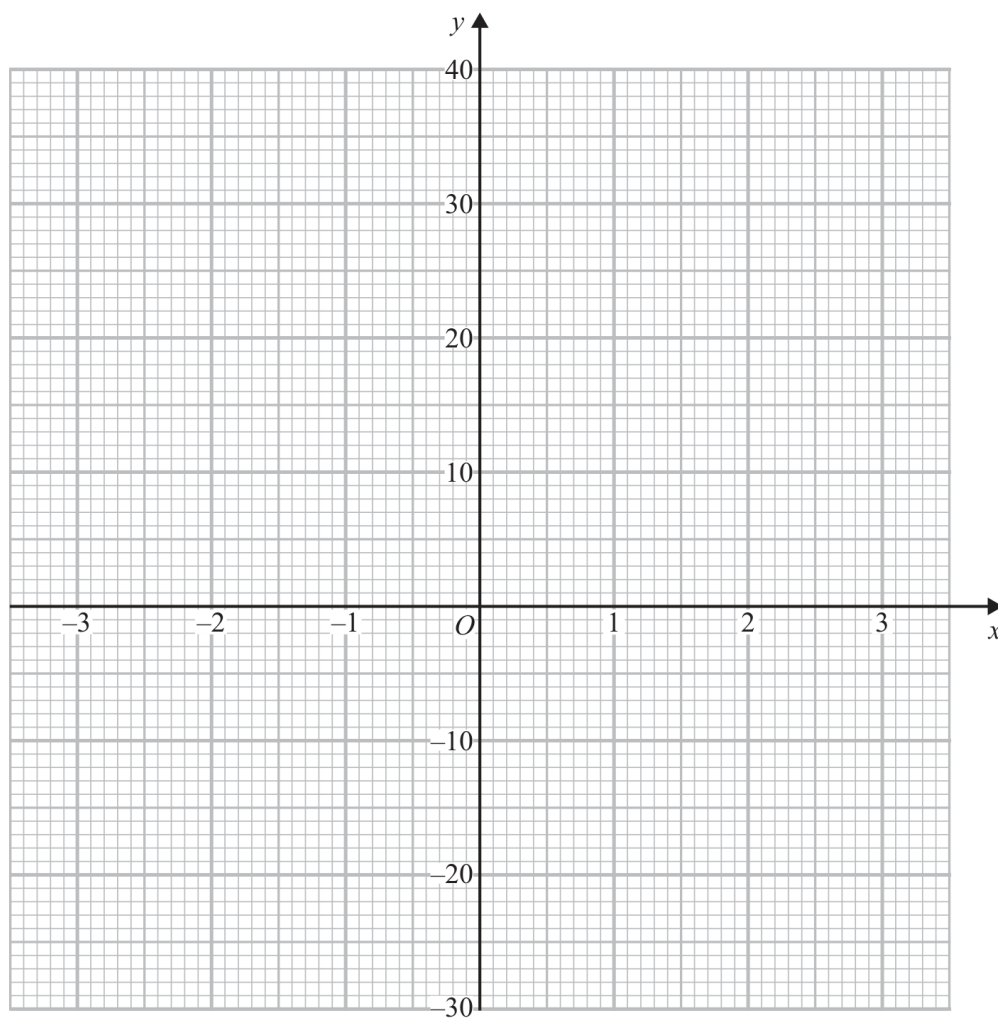
Working Unit 07

4MA1-1H Q10

ORIGINAL QUESTION**3 marks**10 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	-3	-2	-1	0	1	2	3
y		-9	0	3	6		36

(1)

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leq x \leq 3$ 

(2)

Worked Solution 07

Complete and draw a cubic graph

UNIT 1
3 marks**Complete the table**

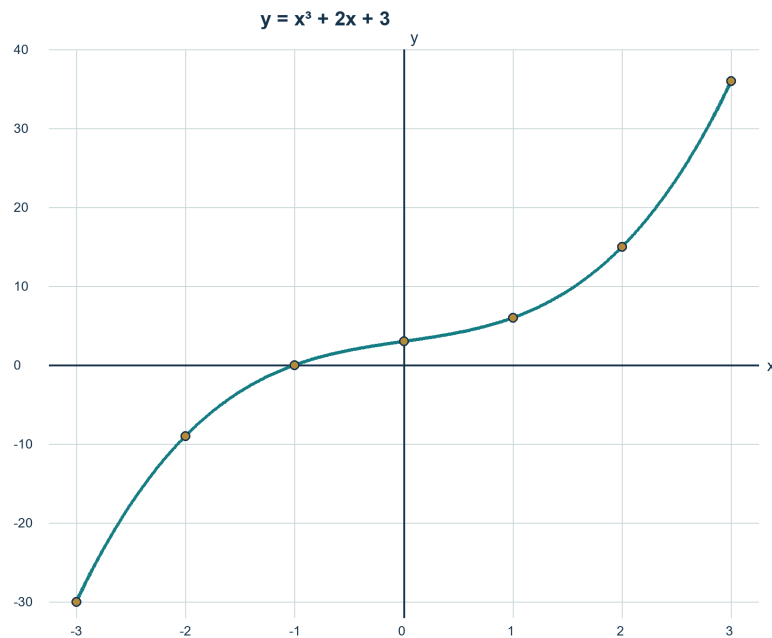
$$y(-3) = -27 - 6 + 3 = -30, \quad y(2) = 8 + 4 + 3 = 15.$$

Plot

$$(-3, -30), (-2, -9), (-1, 0), (0, 3), (1, 6), (2, 15), (3, 36).$$

Join

Draw one smooth increasing cubic curve through all seven points for $-3 \leq x \leq 3$.



Solved reference diagram

Final answer

Missing values -30 and 15 ; smooth cubic through all seven points.

Dependency check The graph uses the completed table; do not use straight-line joins.

Working Unit 08

4MA1-1H Q11

ORIGINAL QUESTION**2 marks**

- 11 $OABC$ is a sector of a circle, centre O

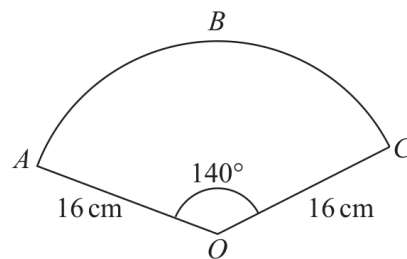


Diagram **NOT**
accurately drawn

$$\text{angle } AOC = 140^\circ$$

$$OA = OC = 16 \text{ cm}$$

Calculate the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 2 marks)

Worked Solution 08

Area of a sector

UNIT 1

2 marks

Use the sector fraction

$$A = \frac{140}{360} \pi (16)^2 = \frac{896\pi}{9} = 312.763 \dots$$

Final answer

$$313 \text{ cm}^2 \text{ (3 s.f.)}$$

Working Unit 09

4MA1-1H Q12

ORIGINAL QUESTION

3 marks

12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

(Total for Question 12 is 3 marks)

Worked Solution 09

Add algebraic fractions

UNIT 1
3 marks**Common denominator**

$$\frac{5}{4} + \frac{x-3}{6x} = \frac{15x}{12x} + \frac{2(x-3)}{12x}.$$

Combine and simplify

$$\frac{15x + 2x - 6}{12x} = \frac{17x - 6}{12x}, \quad x \neq 0.$$

Final answer

$$\frac{17x - 6}{12x}$$

Working Unit 10

4MA1-1H Q14

ORIGINAL QUESTION**4 marks**

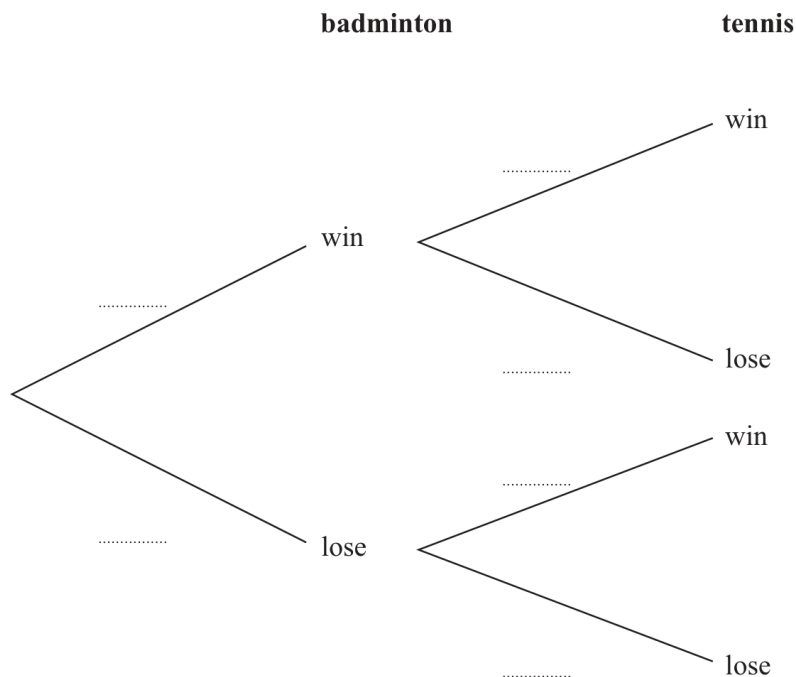
14 Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7

The probability that he will win the game of tennis is 0.4

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Ricardo loses both games.

(2)

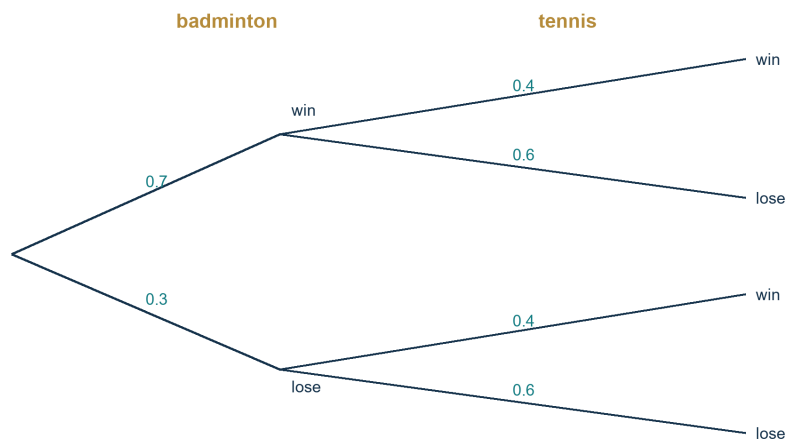
Worked Solution 10

Complete a probability tree

UNIT 1
4 marks**Complementary branches**

$$P(B_W) = 0.7, P(B_L) = 0.3; \quad P(T_W) = 0.4, P(T_L) = 0.6 \text{ after either first result.}$$
Required path

$$P(\text{lose both}) = 0.3 \times 0.6 = 0.18.$$



Solved reference diagram

Final answer

0.18

Dependency check Part (b) reads the two loss branches from the completed tree.

Working Unit 11

4MA1-1H Q15

ORIGINAL QUESTION

4 marks

15 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$k = \dots\dots\dots$
(1)

(b) Show that $\frac{21}{3 - \sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.

Show each stage of your working clearly.

(3)

Worked Solution 11

Powers and rationalising a surd

UNIT 1
4 marks**Part (a)**

$$(\sqrt{3})^5 = (\sqrt{3})^4 \sqrt{3} = 9\sqrt{3} \Rightarrow k = 9.$$

Use the conjugate

$$\frac{21}{3 - \sqrt{2}} \cdot \frac{3 + \sqrt{2}}{3 + \sqrt{2}} = \frac{21(3 + \sqrt{2})}{9 - 2} = 9 + 3\sqrt{2} = 9 + \sqrt{18}.$$

Final answer

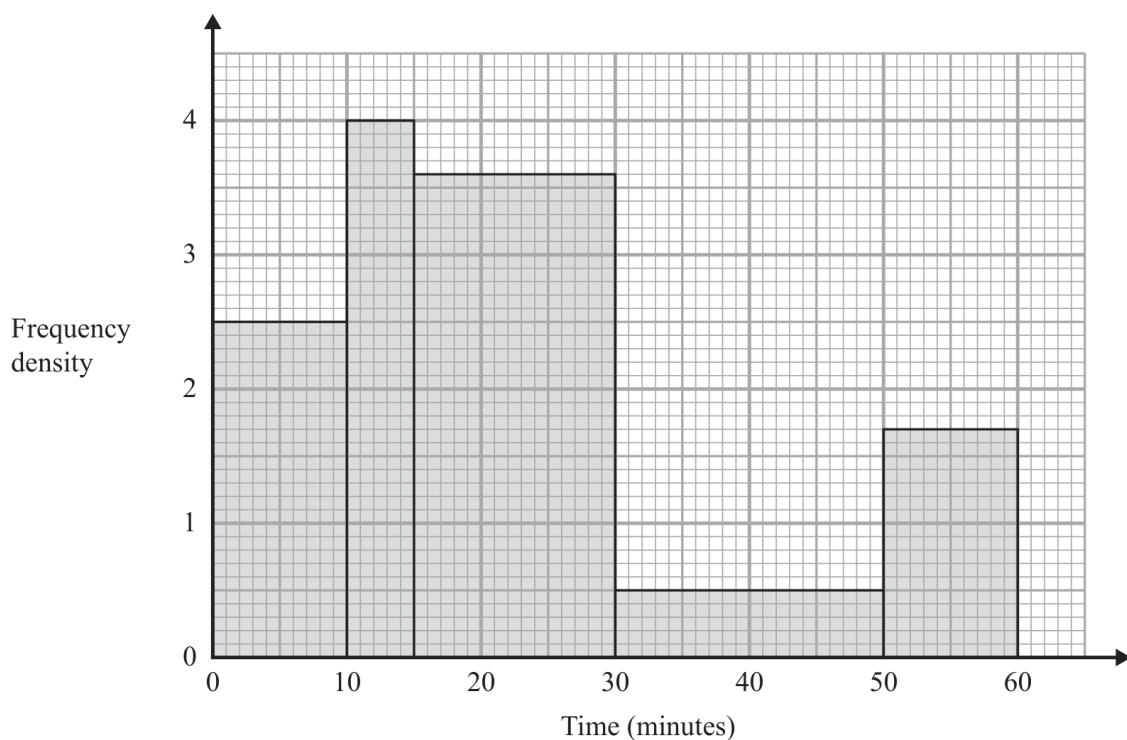
$$k = 9, \quad 9 + \sqrt{18}$$

Working Unit 12

4MA1-1H Q17

ORIGINAL QUESTION**3 marks**

- 17 The histogram shows information about the times, in minutes, that some people were in a shop.



Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

(Total for Question 17 is 3 marks)

Worked Solution 12

Estimate a proportion from a histogram

UNIT 1
3 marks**Find frequencies from areas**

$$10(2.5) = 25, 5(4) = 20, 15(3.6) = 54, 20(0.5) = 10, 10(1.7) = 17.$$

Total frequency

$$25 + 20 + 54 + 10 + 17 = 126.$$

More than 40 minutes

$$10(0.5) + 10(1.7) = 5 + 17 = 22, \quad \frac{22}{126} = \frac{11}{63} \approx 0.175.$$

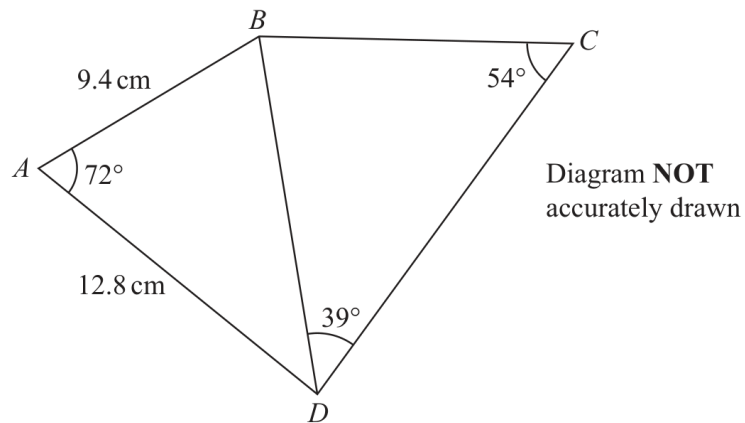
Final answer

$$\frac{11}{63} \approx 0.175$$

Dependency check Only the 40–50 half of the 30–50 class is included.

Working Unit 13

4MA1-1H Q20

ORIGINAL QUESTION**5 marks****20**

Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

Worked Solution 13

Cosine rule followed by sine rule

UNIT 1
5 marks**Find the shared side**

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ, \quad BD = 13.335597 \dots$$

Use the sine rule

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ} \Rightarrow BC = \frac{13.335597 \dots \sin 39^\circ}{\sin 54^\circ} = 10.3735 \dots$$

Final answer

$$BC = 10.4 \text{ cm (3 s.f.)}$$

Dependency check Retain the unrounded value of BD for the second triangle.

Working Unit 14

4MA1-1H Q21

ORIGINAL QUESTION**3 marks**

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....
(Total for Question 21 is 3 marks)

Worked Solution 14

Exactly two counters of the same colour

UNIT 1
3 marks**Use the complement**

$$P(\text{exactly two}) = 1 - P(\text{all same}) - P(\text{all different}).$$

All same

$$P(\text{all same}) = \frac{9 \cdot 8 \cdot 7 + 7 \cdot 6 \cdot 5 + 4 \cdot 3 \cdot 2}{20 \cdot 19 \cdot 18} = \frac{738}{6840}.$$

All different

$$P(\text{all different}) = \frac{6(9 \cdot 7 \cdot 4)}{20 \cdot 19 \cdot 18} = \frac{1512}{6840}.$$

Finish

$$1 - \frac{738 + 1512}{6840} = \frac{4590}{6840} = \frac{51}{76}.$$

Final answer

$$\frac{51}{76}$$

Dependency check Sampling is without replacement; the factor 6 counts the orders of three different colours.

Working Unit 15

4MA1-1H Q25

ORIGINAL QUESTION

5 marks

25 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates $(4, 7)$

R is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points Q and S

Give your answer in the form $ay = bx + c$ where a , b and c are integers.

(Total for Question 25 is 5 marks)

Worked Solution 15

Equation of the other diagonal of a square

UNIT 1
5 marks**Midpoint**

$$M_{PR} = \left(\frac{4+8}{2}, \frac{7-5}{2} \right) = (6, 1).$$

Perpendicular gradient

$$m_{PR} = \frac{-5-7}{8-4} = -3 \Rightarrow m_{QS} = \frac{1}{3}.$$

Equation through the midpoint

$$y - 1 = \frac{1}{3}(x - 6) \Rightarrow y = \frac{1}{3}x - 1 \Rightarrow 3y = x - 3.$$

Final answer

$$3y = x - 3$$

Dependency check Square diagonals bisect each other at right angles.

Working Unit 16

4MA1-2H Q3

ORIGINAL QUESTION

3 marks

3 Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$

(Total for Question 3 is 3 marks)

Worked Solution 16

Subtract mixed fractions

UNIT 1
3 marks**Convert**

$$7\frac{1}{3} - 3\frac{4}{7} = \frac{22}{3} - \frac{25}{7}.$$

Common denominator

$$\frac{154 - 75}{21} = \frac{79}{21} = 3\frac{16}{21}.$$

Final answer

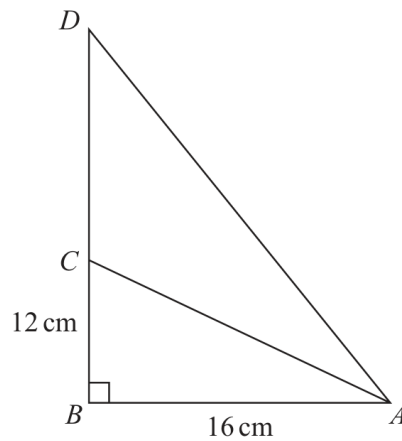
$$3\frac{16}{21}$$

Working Unit 17

4MA1-2H Q10

ORIGINAL QUESTION**5 marks**

10 ABD and ABC are right-angled triangles.



$$AB = 16 \text{ cm} \quad BC = 12 \text{ cm} \quad AD = 1.5 \times AC$$

Find the length of CD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 5 marks)

Worked Solution 17

Connected right-angled triangles

UNIT 1**5 marks****Small triangle**

$$AC = \sqrt{16^2 + 12^2} = 20 \text{ cm.}$$

Large hypotenuse and height

$$AD = 1.5(20) = 30, \quad BD = \sqrt{30^2 - 16^2} = \sqrt{644}.$$

Required segment

$$CD = BD - BC = \sqrt{644} - 12 = 13.3772 \dots$$

Final answer

$$CD = 13.4 \text{ cm (3 s.f.)}$$

Working Unit 18

4MA1-2H Q11

ORIGINAL QUESTION**3 marks**

11 Khalid has a box of counters.

17 of the counters are red

28 of the counters are blue

the rest of the counters are orange

Khalid is going to take at random a counter from the box.

The probability that Khalid will take an orange counter is $\frac{4}{9}$

Work out the number of orange counters that are in the box.

.....
(Total for Question 11 is 3 marks)

Worked Solution 18

Recover a population from a probability

UNIT 1
3 marks**Non-orange counters**

$$17 + 28 = 45, \quad P(\text{non-orange}) = 1 - \frac{4}{9} = \frac{5}{9}.$$

Total and orange

$$N = 45 \div \frac{5}{9} = 81, \quad 81 - 45 = 36.$$

Final answer

36 orange counters

Working Unit 19

4MA1-2H Q12

ORIGINAL QUESTION

3 marks

12 Expand and simplify $3x(2x+5)(7x-4)$

(Total for Question 12 is 3 marks)

Worked Solution 19

Expand three algebraic factors

UNIT 1
3 marks**First two brackets**

$$(2x + 5)(7x - 4) = 14x^2 + 27x - 20.$$

Multiply by 3x

$$3x(14x^2 + 27x - 20) = 42x^3 + 81x^2 - 60x.$$

Final answer

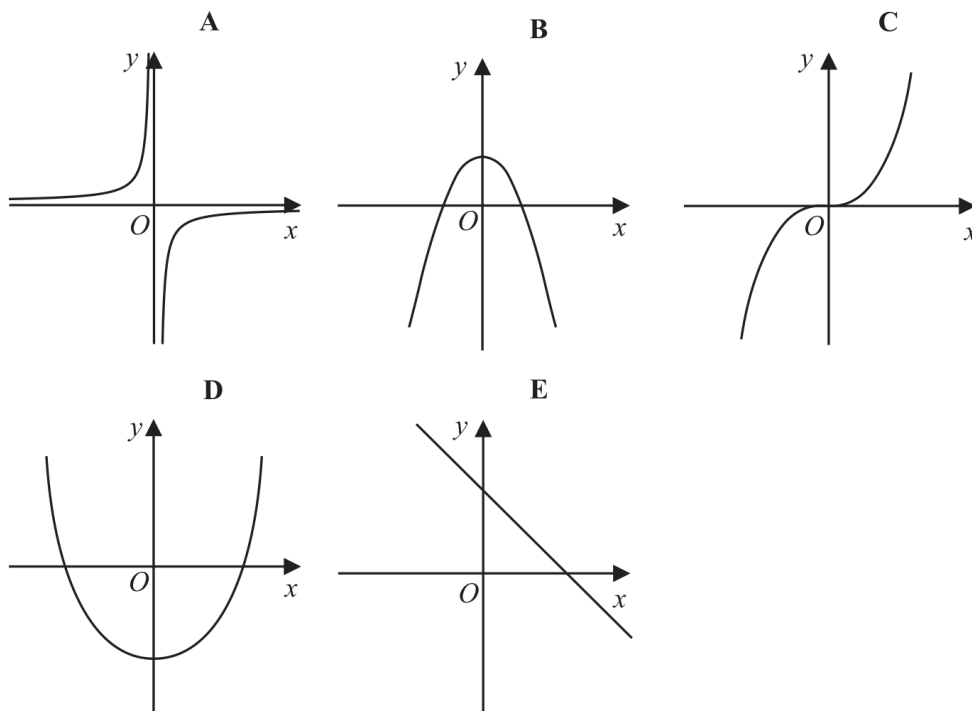
$$42x^3 + 81x^2 - 60x$$

Working Unit 20

4MA1-2H Q14

ORIGINAL QUESTION**2 marks**

14 Here are five graphs.



(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$

.....
(1)

(b) Write down the letter of the graph that could have the equation $y = 2x^3$

.....
(1)

Worked Solution 20

Recognise graphs from equations

UNIT 1
2 marks**Part (a)**

$y = 5 - x^2$ is a downward parabola with positive y -intercept: graph B.

Part (b)

$y = 2x^3$ is an increasing cubic through the origin: graph C.

Final answer

(a) B (b) C

Working Unit 21

4MA1-2H Q15

ORIGINAL QUESTION

3 marks

15 Simplify fully $\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$

.....
(Total for Question 15 is 3 marks)

Worked Solution 21

Negative power of an algebraic fraction

UNIT 1
3 marks**Simplify inside**

$$\frac{2a^3}{10a^7c^2} = \frac{1}{5a^4c^2}.$$

Use the negative power

$$\left(\frac{1}{5a^4c^2}\right)^{-3} = (5a^4c^2)^3 = 125a^{12}c^6.$$

Final answer

$$125a^{12}c^6$$

Working Unit 22

4MA1-2H Q18

ORIGINAL QUESTION**2 marks**

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

Worked Solution 22

Convert a recurring decimal to a fraction

UNIT 1
2 marks**Align a complete repeating block**

$$x = 0.\overline{702}, \quad 1000x = 702.\overline{702}.$$

Subtract and simplify

$$999x = 702 \Rightarrow x = \frac{702}{999} = \frac{26}{37}.$$

Final answer

$$\frac{26}{37}$$

Working Unit 23

4MA1-2H Q20

ORIGINAL QUESTION**6 marks**

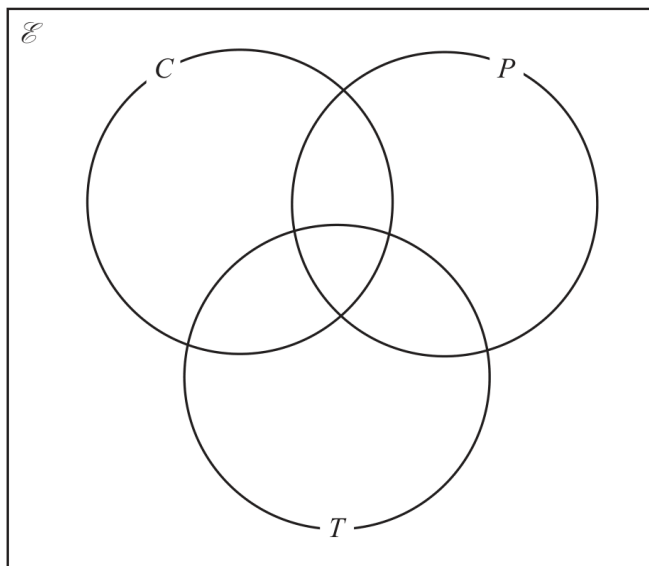
- 20** 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

- 43 grow carrots
- 12 grow carrots and potatoes and tomatoes
- 18 grow carrots and potatoes
- 27 grow carrots and tomatoes
- 32 grow potatoes and tomatoes
- 29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

- (a) Complete the Venn diagram to show this information.



(3)

- (b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

- (c) Calculate the probability that this gardener also grows potatoes.

(2)

Worked Solution 23

Three-set Venn diagram and conditional probability

UNIT 1
6 marks**Intersections**

$$C \cap P \cap T = 12, CP \text{ only} = 6, CT \text{ only} = 15, PT \text{ only} = 20.$$

Single-set regions

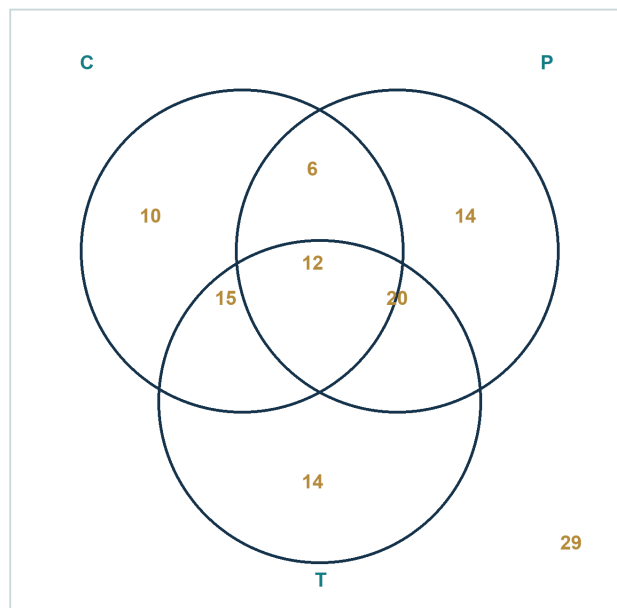
$$C \text{ only} = 43 - (6 + 15 + 12) = 10. \text{ If } P \text{ only} = T \text{ only} = k, 10 + 6 + 15 + 12 + 20 + 2k = 120 - 29, \text{ so } k = 14.$$

Part (b)

$$n(T' \cap C) = 10 + 6 = 16.$$

Part (c)

$$P(P \mid C) = \frac{n(P \cap C)}{n(C)} = \frac{18}{43}.$$



Solved reference diagram

Final answer

Regions (10, 14, 14, 6, 15, 20, 12; outside 29), 16, $\frac{18}{43}$

Dependency check Parts (b) and (c) read the completed diagram; the diagram appears once.

Working Unit 24

4MA1-2H Q22

ORIGINAL QUESTION**4 marks**

22 The diagram shows a square inside rectangle $ABCD$

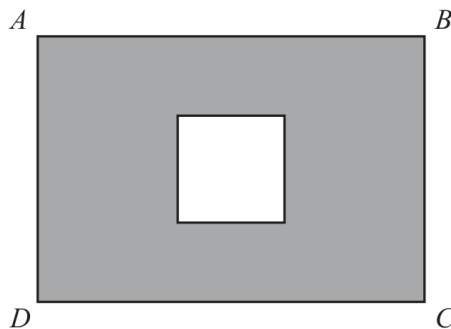


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X\text{cm}^2$

$AB = 11.5\text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2\text{ cm}$ correct to 2 significant figures

side of square $= 4.1\text{ cm}$ correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

Worked Solution 24

Bounds for a shaded area

UNIT 1
4 marks**Input bounds**

$$11.25 \leq AB < 11.75, \quad 9.15 \leq BC < 9.25, \quad 4.05 \leq s < 4.15.$$

Upper shaded area

$$X_U = 11.75(9.25) - 4.05^2 = 92.285.$$

Lower shaded area

$$X_L = 11.25(9.15) - 4.15^2 = 85.715.$$

Choose the safe accuracy $85.715 \leq X < 92.285$; both bounds round to 90 to 1 significant figure.**Final answer**

$$X = 90 \text{ cm}^2 \text{ (1 s.f.)}$$

Working Unit 25

4MA1-2H Q23

ORIGINAL QUESTION**4 marks**

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$$p = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

Worked Solution 25

Simplify a division of algebraic fractions

UNIT 1
4 marks**Factorise**

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)}.$$

Multiply by the reciprocal and cancel

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \times \frac{10x(x+6)}{x(x+5)} = \frac{4 \cdot 10}{5} = 8.$$

Final answer

$$p = 8$$

Working Unit 26

4MA1-2H Q24

ORIGINAL QUESTION**4 marks**

24 The diagram shows a square-based pyramid $ABCDE$

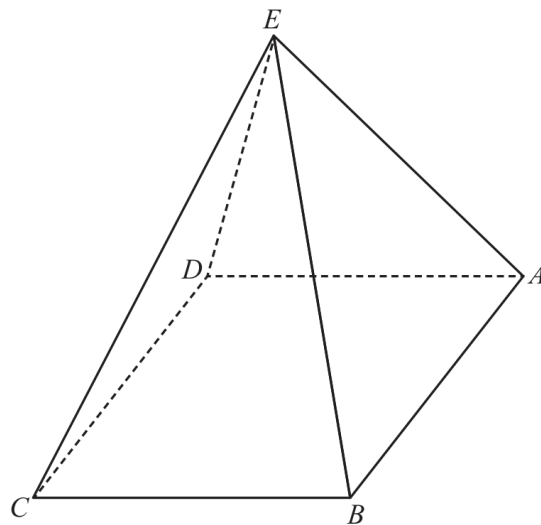


Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

M is the centre of the horizontal square base $ABCD$

Q is the midpoint of AB

$$\text{Angle } EQM = 80^\circ$$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

Worked Solution 26

Three-dimensional trigonometry in a pyramid

UNIT 1
4 marks**Choose a scale**

Let $AB = 1$. Then $QM = \frac{1}{2}$ and $AM = \sqrt{(1/2)^2 + (1/2)^2} = \frac{1}{\sqrt{2}}$.

Vertical height

$$EM = QM \tan 80^\circ = \frac{1}{2} \tan 80^\circ.$$

Sloping edge

$$EA = \sqrt{EM^2 + AM^2} = \sqrt{\frac{1}{4} \tan^2 80^\circ + \frac{1}{2}} = 2.922474 \dots$$

Final answer

$$n = 2.92 \text{ (3 s.f.)}$$

Working Unit 27

4MA1-2H Q25

ORIGINAL QUESTION

3 marks

25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a , b and c are integers.

.....
(3)

Worked Solution 27

Complete the square

UNIT 1**3 marks****Rearrange**

$$28 + 24x - 6x^2 = 28 - 6(x^2 - 4x).$$

Complete

$$28 - 6((x - 2)^2 - 4) = 52 - 6(x - 2)^2.$$

Final answer

$$52 - 6(x - 2)^2 \quad (a = 52, b = 6, c = 2)$$

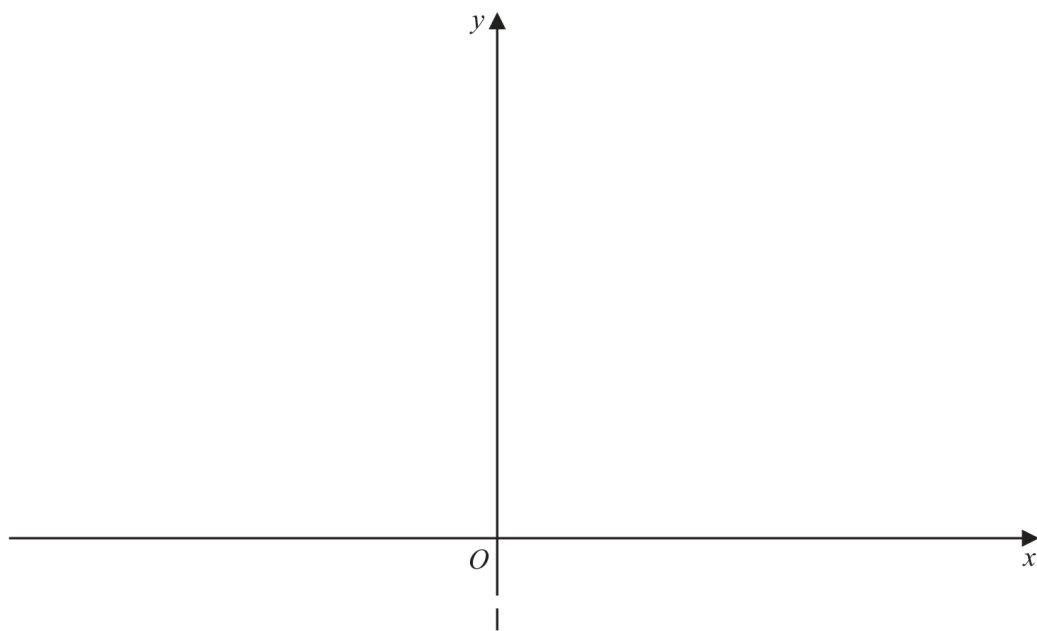
Working Unit 28

4MA1-2H Q25

ORIGINAL QUESTION**3 marks**

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

Worked Solution 28

Sketch a quadratic from completed-square form

UNIT 1
3 marks**Turning point**

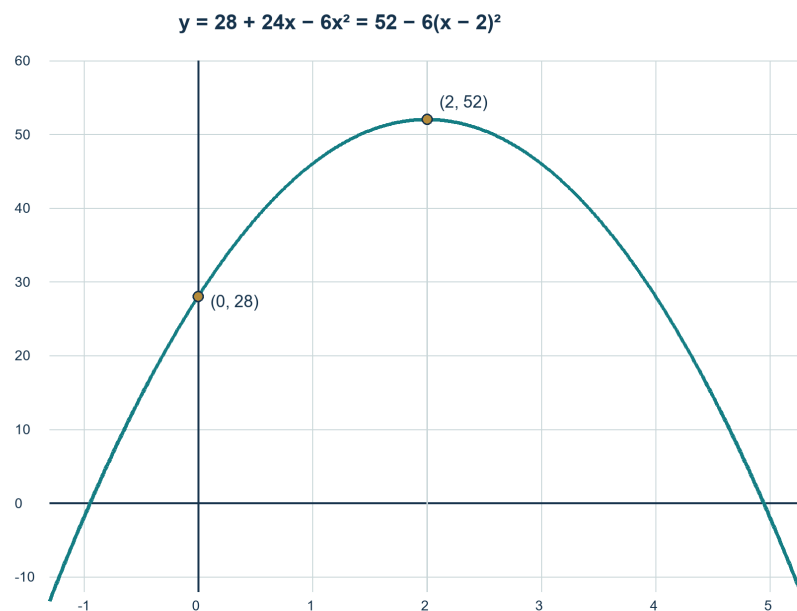
$$y = 52 - 6(x - 2)^2 \Rightarrow \text{maximum turning point } (2, 52).$$

y-intercept

$$x = 0 \Rightarrow y = 28, \text{ so the graph crosses the } y\text{-axis at } (0, 28).$$

Shape

The coefficient of $(x - 2)^2$ is negative, so draw a symmetric downward parabola about $x = 2$.



Solved reference diagram

Final answer

Downward parabola with turning point $(2, 52)$ and y -intercept $(0, 28)$.

Dependency check Sequenced immediately after WU027; the printed equation is repeated, so the crop is self-contained.